

ROY CARLOUS CHRISTUDASS

AI Software Engineer

Dayton, OH (open to relocation) · roy4edu@gmail.com · (326) 467-1939 ·  ·  · Portfolio

Technical Skills

Languages: Python, Java, C++, C#/.NET, JavaScript, SQL, HTML/CSS

AI and ML: scikit-learn, embeddings and semantic search, hybrid retrieval (BM25 + dense), retrieval evaluation, LLM integration (Ollama, Gemini API), OpenCV, MediaPipe, Tree-sitter

Engineering: FastAPI, Spring Boot, REST API design, message queues, fault tolerance (retries, dead-letter queues, idempotency), React, Vite, Tailwind CSS, AWS (SQS, S3, DynamoDB), Docker, PostgreSQL, MySQL, Git, pytest, JUnit

Test and embedded systems: Teradyne IG-XL (UltraFLEX, UltraFLEXplus), instrument drivers, WinDbg and JetBrains profilers, acceptance test automation, Arduino, serial protocols

Data and BI: Power BI dashboards, pandas, NumPy, ETL pipelines, KPI and trend reporting

Professional Experience

HCLTech (contractor to Teradyne Inc.)

Jan 2022 – Aug 2024

Senior Software Engineer (Jan 2024) · Software Engineer (Aug 2022) · Graduate Engineer Trainee

Chennai, India

- Owned C++ and C#/.NET instrument driver development for Teradyne's IG-XL automated test platform, running on UltraFLEX and UltraFLEXplus testers in semiconductor fabs worldwide, where a driver defect stops a production line. Promoted twice in under three years.
- Was the sole escalation point for critical stopper issues affecting Teradyne's end customers, and built the Power BI dashboards that tracked issue trends across both tester platforms. Scattered escalation records became a view of where defects clustered by platform, module and instrument, cutting mean time to resolution on the recurring classes.
- Cut root-cause time on memory leaks and performance regressions by moving legacy diagnostic workflows onto an AI-assisted debugging framework, using WinDbg, JetBrains Timeline Profiler and automated flagging of regressions between builds, which stopped defects that had kept coming back from being re-escalated.
- Worked across teams migrating the IG.NET framework from C++ to a modern C#/.NET architecture, triaging the defect backlog across the ported modules and profiling runtime performance against the original build to catch bottlenecks before deployment. Zero production stoppers on the releases managed.
- Triaged and resolved 150+ defects across three analog driver codebases (DC30, DC70, DC75) by tracing failures through automated acceptance-test logs and captured measurement data, shipped new language nodes extending automated test coverage to next-generation hardware, and built and repaired the unit and acceptance suites behind all three.

Projects

DiffLens

Demo · Code

Python, FastAPI, scikit-learn, Tree-sitter, PostgreSQL, React, Docker, Ollama

- Built a code review service that parses Python and Java into Tree-sitter ASTs to measure cyclomatic complexity and nesting depth, scores pull-request risk from those features alongside finding severity and diff size, and surfaces recurring issues by TF-IDF similarity against past findings.
- Added a local LLM pass turning findings into review comments, wired GitHub webhooks so every pull request analyzes itself and posts inline, then shipped it as a four-container Docker Compose stack behind a 14-file pytest suite.

TaskForge

Demo · Code

Java 17, Spring Boot, AWS (SQS, DynamoDB, S3), Docker, LocalStack, React

- Built a distributed job-processing system: a REST API enqueues to SQS, independent workers process and deliver to S3 behind presigned URLs, and a React dashboard polls live job state and queue depth. Engineered for failure rather than the happy path: exponential backoff with 20% jitter, a dead-letter queue after three attempts, idempotency keys returning 409, SIGTERM graceful drain, and correlation IDs through every hop so any failed job stays traceable.

GestureControl

Demo · Code

Python, OpenCV, MediaPipe, PyFirmata, Arduino UNO

- Built a 30 FPS vision-to-actuator control loop: MediaPipe's two-stage model, an SSD palm detector then 21-point 3D landmark regression, feeds a geometric classifier that reads finger state from tip-versus-joint positions and drives a 4-channel relay module over serial. A 15px jitter threshold and 3-frame stabilization window suppress false triggers, and a simulation mode tests the whole loop with no board attached.

CodeAtlas

Case study · Code

Python, sentence-transformers, Tree-sitter, BM25, NumPy

- Built semantic search over a 152-file codebase using Tree-sitter chunking on declaration boundaries with hybrid BM25 and embedding retrieval, then wrote the evaluation harness that scores it: 36 labelled questions across four retrieval and three chunking strategies, reaching recall@5 0.86. Reported the results that went against expectation, including a standard cross-encoder reranker that made ranking worse on code.

Education

M.S. Computer Science, Wright State University, Fairborn, OH

Aug 2024 – Aug 2026

Coursework: Foundations of AI, Information Retrieval, Algorithm Design and Analysis, Distributed Computing, Advanced Computer Networks, Reverse Engineering and Program Analysis.

B.E. Electronics and Communication Engineering, Sathyabama Institute of Science & Technology

Aug 2018 – Jun 2022